

Sai Kiran

Full-Stack Developer

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Summary

Software Engineer with **3+ years of experience** architecting, modernizing, and scaling full-stack applications across **React, Next.js, Python, and AWS ecosystems**. Proven ability to design **secure, data-intensive, and high-availability systems** that power mission-critical business workflows. Experienced in **technical strategy, modernization of legacy stacks**, and driving architectural excellence through mentorship, automation, and deep system understanding. Skilled in **frontend performance optimization, backend API design, and cloud-native DevOps practices** using **EKS, DynamoDB, and CI/CD automation**. Known for leading large-scale initiatives from concept to deployment, improving performance, maintainability, and developer velocity while fostering a **security-first, reliability-driven engineering culture**.

Work Experience

Software Development Engineer, Aug 2023 - Present

Universal Studios, Atlanta

- **Led architecture and modernization efforts** for a distributed UI SDK platform serving Android, iOS, and Web clients, transitioning from legacy REST APIs to a **modular, event-driven Python microservice** model.
- Designed and deployed **Next.js and React-based operational dashboards**, improving real-time visibility for logistics and merchant operations teams.
- Architected **secure RBAC and audit logging mechanisms** using **Auth0 and AWS KMS**, ensuring compliance with internal security standards.
- Implemented **data caching and async processing** pipelines using **Redis + AWS SQS**, improving backend response times by **35%**.
- Drove **frontend modernization strategy**, introducing Next.js Server Components and shared UI libraries for consistency across business portals.
- Mentored 3 junior developers through **code reviews, architectural sessions, and system design walkthroughs**, accelerating delivery velocity.
- Integrated **observability pipelines** (Datadog + CloudWatch) with detailed metrics and runbooks, reducing production incident resolution time by **40%**.
- Collaborated with product and UX teams to deliver **data-intensive tools** used by Customer Service and Fraud Operations, improving workflow efficiency.

Full-Stack Engineer, Jan 2021 - Aug 2022

Capgemini, Bengaluru, India

- Engineered **scalable full-stack web applications** using **React, TypeScript, and Python (FastAPI)** to automate business operations.
- Led **legacy PHP monolith refactor** into modular microservices, enabling 50% faster deployment cycles and reducing code complexity.
- Built **GraphQL APIs and high-performance SQL queries** (PostgreSQL) optimized for analytical workloads and dashboard visualizations.
- Designed **automated CI/CD pipelines** with Docker and GitLab CI, reducing manual release time by **70%**.
- Implemented **secure data transmission and encryption layers** ensuring compliance with privacy and audit requirements.
- Partnered with QA and DevOps teams to automate **end-to-end tests** using **Playwright** and containerized environments.
- Enhanced observability and uptime through structured logging and health monitoring with **Prometheus and Grafana**.

Skills

Frontend: React 19, Next.js 15, TypeScript, Redux/Zustand, Tailwind, Playwright (E2E)

Backend: Python (FastAPI, Flask), PHP (Zend), Node.js, REST & GraphQL APIs

System Design: High Availability, Data Modeling, Caching (Redis), RBAC, Scalability

Databases: Oracle, PostgreSQL, DynamoDB, Redis, Elasticsearch

Cloud & DevOps: AWS (EKS, Lambda, SQS, SNS, CloudWatch), Docker, Kubernetes, ArgoCD, GitLab CI

Observability: Datadog, Splunk, Prometheus, Grafana, Logging & Alerting Frameworks

Security: Encryption at Rest/Transit, Auth0/Okta, OAuth2, Audit Trails

Leadership: Technical Mentorship, Architecture Reviews, Modernization Strategy, Cross-Functional Collaboration

Projects

Platform Modernization & System Migration (Legacy to Next.js + Python)

Stack: Next.js, Python (FastAPI), AWS EKS, PostgreSQL, Auth0

- Spearheaded migration from **PHP Zend framework** to a **modern React + Python microservice architecture**, improving maintainability and performance.
- Introduced **modular system design patterns** that reduced dependency cycles and simplified onboarding for new engineers.
- Integrated **fine-grained RBAC and encrypted storage**, improving data security and auditability across systems.

Data-Intensive Analytics Dashboard for Business Operations

Stack: React, TypeScript, DynamoDB, AWS Lambda, CloudWatch

- Developed a data visualization platform used by finance and operations teams for real-time insights.
- Implemented **ETL pipelines** and API layers handling millions of records daily, achieving 99.9% uptime.
- Created dynamic caching and pagination mechanisms that improved query performance by **45%**.

Education

Master's Degree In Computer Science, Aug 2022 - Dec 2023

Kent State University, Kent, Ohio

Coursework: Big Data Analytics, Multicore Computing, Digital Forensics, Advanced Computer Graphics, Algorithms, Information Security, Advance Database System Design, Security of Cloud and Distributed System.

Bachelor's Degree in Computer Science, Jun 2018 - Mar 2021

Loyola Academy UG and PG College, Hyderabad, India

Certifications

AWS Certified Developer – Associate (2025)

Certified Scrum Developer (CSD) - Agile Alliance